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MILITARY SPECIFICATION

OVENS, BAKING AND ROASTING, FORCED CONVECTION, GAS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one and two compartment stationary gas fired, forced convection baking and roasting ovens.

* 1.2 Classification. Ovens shall be of the following types and styles as specified (see 6.2).

Type I - One compartment
Style A - With glass door
Style B - Without glass door

Type II - Two compartment
Style A - With glass door
Style B - Without glass door

Style 1 - (see 6.4)
Model B - (see 6.4)

2. APPLICABLE DOCUMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-2316 - Bakery Equipment, Including Unit, Assemblies, Repair Parts, and Tools, Preparation for Delivery Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN GAS ASSOCIATION, (A.G.A.)

Directory of Certified Appliances and Accessories

(Application for copies should be addressed to the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z 21.41 - Quick Disconnect Device for Use with Gas Fuel

Z 21.45 - Flexible Connectors of Other than All-Metal Construction for Gas Appliances

Z 83.12 - Gas Food Service Equipment - Baking and Roasting Ovens

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

A 176 - Stainless and Heat-Resisting Chromium Steel Plate Sheet, and Strip

A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality

A 412 - Stainless and Heat-Resisting Chromium-Nickel-Manganese Steel Plate, Sheet, and Strip

A 463 - Steel Sheet, Cold-Rolled, Aluminum-Coated Type 1

A 569 - Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled-Sheet and Strip, Commercial Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Rd., P.O. Box 1468, Ann Arbor, MI 48106.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard product. The oven shall, as a minimum, be in accordance with the requirements of this document and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this document. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in

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accordance with this document shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the government consistent with their normal commercial practices.

3.2 Codes and standards. The convection ovens shall conform to the applicable requirements of NSF Standard No. 4, and ANSI Standards Z 21.41, Z 21.45, and Z 83.12. Satisfactory evidence that these requirements have been met shall be submitted to the contracting officer or his authorized representative prior to the start of production (see 4.3).

3.3 Materials (see 6.3) and components. Materials and components shall be as specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer for his standard commercial convection oven, provided the completed item complies with all provisions of this document.

3.3.1 Steel.

3.3.1.1 Stainless and heat-resisting steel. The stainless and heat-resisting steel shall conform to type 202 of ASTM A 412; type 302 or 304 of ASTM A 167; or any of 400 types of ASTM A 176.

3.3.1.2 Carbon steel. Carbon steel shall be commercial quality (CQ) conforming to ASTM A 366 or ASTM A 569 with an AISI steel designation of 1010-1020.

3.3.1.3 Aluminum-coated steel. The aluminum-coated steel shall conform to type I of ASTM A 463.

3.4 Design. Convection ovens shall be designed for baking and roasting and shall incorporate a two-speed oven blower to provide for circulation of forced air throughout the oven compartment. Type I and II ovens shall have a single oven compartment and two oven compartments, respectively. The type I oven shall consist essentially of a body, oven compartment linings, doors with handle(s), shelves, shelf support, burner units, two-speed oven blower, temperature control, timer, oven light, burner on-off control, blower speed selector switch (low and high), and flue riser. The type II oven shall consist of two type I ovens for stacking and securing above one another. Style A ovens shall have heat tempered glass viewing doors. Each oven compartment shall be capable of being operated independently and shall be individually controlled by a thermostat. The convection oven hourly gas input rating shall not exceed 115,000 British thermal units (Btu) for type I ovens and 230,000 Btu for type II ovens.

3.4.1 Mounting. The type I oven shall be mounted on legs directly attached to the four bottom corners of the oven or mounted on an open or enclosed stand. The type II oven shall be mounted on legs directly attached to the

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four bottom corners of the oven. All ovens shall be provided with leveling devices.

3.4.1.1 Legs and stands. All legs shall be fabricated of stainless steel. All legs except on the open stand shall be removable. Open stands and the front, bottom and side panels of the enclosed stands shall be fabricated of stainless steel. Open and enclosed stands shall be reinforced and their extremities shall not protrude beyond the extremities of the oven exterior.

3.4.2 Dimensions. Convection ovens shall be within the dimensional limits shown in table I.

TABLE I. Dimensions (inches)

Measurements	Minimum	Maximum
Width overall	33	52
Depth overall	30	48
Height 1/ overall (excluding vent):		
Type I	54	65
Type II	57	78
Height 1/ to bottom deck:		
Type I	30	38
Type II	11	19-3/4
Height 1/ to top deck (type II)	30	53
Compartment (inside usable):		
Width	26-1/2	29-1/4
Depth	20	30
Height	19-3/8	28

1/ Height from floor, including supporting legs and base.

NOTE: When the blower motor is not an integral part of the oven, the motor encasement shall not extend beyond 9 inches on the side opposite the oven controls.

3.5 Construction. Type I and II oven components fabricated of stainless steel, carbon steel, or aluminized steel shall conform to 3.3.1.1, 3.3.1.2, or 3.3.1.3, respectively.

3.5.1 Body. The body shall be of a frame or shell construction. When frame construction is used, the frame shall consist of a base frame, top frame, and vertical angles to form a rigid assembly. Exterior sheets shall be securely fastened to the frame, enclosing the frame construction therein. When of shell construction, the outer shell shall be stiffened and reinforced. The front of the ovens shall be provided with means for mounting the controls.

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3.5.1.1 Type I. The exterior surfaces of the type I oven except the bottom and back shall be fabricated of stainless steel. The bottom and back surface may be fabricated of carbon steel or aluminized steel.

3.5.1.2 Type II. The exterior surfaces of the type II oven (front, sides, and top) shall be fabricated of stainless steel. The bottom, top, and back of the lower oven, and the bottom and back of the upper oven shall be fabricated from carbon steel or aluminized steel. The top inside surface of the upper oven may be fabricated of carbon steel provided the exterior surface is stainless steel.

3.5.2 Oven compartment linings. Each oven compartment shall be provided with oven compartment linings fabricated from stainless steel specified in 3.3.1.1 or aluminized coated steel specified in 3.3.1.3 or porcelain enamel over carbon steel specified in 3.3.1.2. The oven compartment linings shall not have open joints and shall be reinforced.

3.5.2.1 Vent control or air baffle. Vent control or air baffles when specified (see 6.2), shall be fabricated from stainless steel (3.3.1.1), aluminum-coated steel (3.3.1.3) or porcelain enamel over carbon steel (3.3.1.2). The vent control or air baffle shall be manually operated and provided on each oven compartment.

3.5.3 Doors. The doors shall be of the vertical split, side opening type. Vertically split doors when in the fully open position shall open flush with the oven interior sidewalls and shall not obstruct the removal of the shelves and shelf supports. Handle(s) shall be provided on the doors. When one handle is provided on a convection oven with two doors, both doors shall operate simultaneously when opened and may close simultaneously. Heat-resisting viewing window(s) shall be located in the style A oven doors so that at least the center three shelves may be viewed when the doors are closed. Style B oven doors shall be without windows. Each convection oven shall be provided with a latch, clip, or other means to hold the doors in the closed position. The oven door interior shall be fabricated from stainless steel. When specified (see 6.2), the oven door interior shall be fabricated from aluminized steel, or shall be porcelain enamel on carbon steel. The convection oven may be provided with a door control switch.

3.5.4 Shelves. Each oven compartment shall be provided with a minimum of five equally spaced removable shelves supported by removable shelf supports. Shelves shall be fabricated of steel rod, steel frame with steel slats, or expanded steel. Shelves shall not jam or buckle when placed into, or removed from, the shelf supports. Each shelf shall be capable of supporting a load of 20 pounds per square foot without more than 3/16 inch deflection, dislodgement, or permanent deformation when tested as specified in 4.4.1.

3.5.5 Venting. A means shall be furnished for venting compartments. When a flue riser is provided, it shall be located at the rear of the oven and a

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single flue shall vent both the upper and lower oven compartments on type II ovens.

3.5.6 Fuel system. The fuel system, including burners, shall be designed to efficiently burn natural gas or liquified petroleum gas (LPG) with appropriate conversion, as specified (see 6.2).

3.5.6.1 Natural gas. The unit shall be furnished to operate on natural gas. Main burners shall be provided with fixed orifices with orifice hoods or spuds that can be interchanged. The orifice hoods or spuds installed for natural gas shall be unpainted brass. An automatic pilot including a 100 percent shut-off shall be provided on each unit.

3.5.6.2 LPG. The contractor shall furnish black color-coded orifice hoods or spuds to convert the unit to operate on the type of LPG specified (see 6.2). The contractor shall also furnish complete instructions for conversions to operate on LPG.

3.5.6.3 Automatic ignition. All main burners shall be equipped with means for automatic ignition of the gas. A pilot, including a 100 percent shut-off device shall be provided for each oven burner. The pilot shall automatically shut off the gas to all oven burners including oven pilot burners, in the event that means of ignition becomes inoperative.

3.5.6.4 Burner unit(s). Each oven compartment shall be provided with independently operated burner unit(s) located to provide heat in the oven compartment. The burner unit(s) shall be fully protected and removable from the front.

3.5.6.5 Accessories. A pressure regulator for natural gas shall be furnished with each unit. The unit shall be furnished with a connector conforming to Z 21.45 and a quick disconnect conforming to Z 21.41. The connector length shall be 60 inches.

3.5.7 Electrical characteristics. Unless otherwise specified (see 6.2), the electrical controls, lights, switches and blower shall be suitable for operation on nominal 120-volt, 60-hertz, 1-phase (ac) circuit. All controls, switch(es), and indicator light(s) shall be located in a control panel at the front of the oven.

3.5.7.1 Oven blower. Each oven compartment shall have a two speed blower. The oven blower shall provide even circulation and heat distribution throughout the oven compartment such that no mean shelf temperature differs from any other mean shelf temperature by more than 25°F when tested as specified in 4.4.5. The oven blower shall operate whenever the door(s) are closed. A separate oven blower control switch may be provided for cooling. The oven blower motor shall be of the capacitor or split phase type. The oven blower shall be adequate to sustain continuous operation at high and low oven blower motor speed settings. Motor speed on the high speed setting shall be

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nominal 1725 rpm and the low speed setting shall be between 750 and 1125 rpm. Switching from one speed to the other shall be accomplished without damage to the motor and to the motor electrical system. A protective device shall be provided over the blower to prevent the roasting pan handle, when in the raised position, from hitting the blower. The oven blower wheel shall be fabricated of stainless steel as specified in 3.3.1.1 or other material suitable for the purpose intended. If stainless steel is not used, the blower wheel shall be permanently coated with a material to prohibit corrosion.

3.5.7.2 Oven light. The oven shall be provided with light(s) to illuminate the oven interior. The light(s) shall be recessed or protected with a heated tempered glass window(s). The oven light(s) may be actuated by a manual switch or automatically by a door switch. The light and protective window shall be accessible for light replacement and window cleaning.

3.5.7.3 Indicator light. Ovens shall be furnished with a separate indicator light that glows when the oven is operating and goes out whenever the proper temperature has been attained.

3.5.7.4 Operation. Ovens shall not leak gas and burners, controls, and moving parts shall operate and perform as intended when tested as specified in 4.4.2.

3.5.8 Temperature control. Each oven compartment shall have a manually operated thermostat to provide automatic heat control of the oven compartment and permit cooking operations to be accomplished by maintained temperature. The thermostat shall be capable of maintaining any oven temperature within the range of 250°F to at least 400°F, plus or minus 13°F of the temperature setting when tested as specified in 4.4.3. The thermostat actuating components shall be protected against moisture, grease, and direct heat radiation. An "OFF" position shall be marked on the thermostat dial or a separate switch shall be provided. The temperature control dial shall be marked from 150°F to at least 500°F.

3.5.8.1 Preheat time. The oven shall attain a compartment temperature of 400°F within 20 minutes in an ambient temperature of 70°F plus or minus 5°F when tested as specified in 4.4.4.

3.5.9 Oven timer. Each oven compartment shall be provided with a 5 hour oven timer installed in the control panel at the oven front. The oven timer shall be equipped with an alarm signal which will sound when the preset time has been completed.

3.5.10 Terminal board or box. A terminal board or box shall be furnished on all ovens. The board or box shall be within the oven enclosure and shall provide for permanent connection to the installation wiring system. The terminal board or box shall have a grounding terminal with connection to the oven outer body.

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3.6 Marking.

3.6.1 Data-name plates. The ovens shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to the oven by rivets, screws, or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address and Telephone Number
Supplier's Name, Address and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number and Serial Number
Government Approved Manual Number (see 6.2).

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the oven.

3.6.2 Users' instruction plate. Each type and style of oven shall be equipped with a users' instruction plate, permanently fastened to the outside front of the unit. The plate shall be constructed of corrosion-resistant metal having a minimum thickness of 0.0239 inch, and shall have the following information etched or stamped thereon in legible characters. Step 3 (as shown below) may be omitted if the unit does not have valves. The word "CONVECTION OVEN" shall be in characters not less than 3/16 inch high.

CONVECTION OVEN

USERS' INSTRUCTIONS

1. Turn gas burner controls on and adjust thermostat to desired oven temperature.

2. Oven. If oven burner fails to light, due to failure of pilot flame, manually shut gas off and wait 5 minutes with combustion compartment door open, then reignite the automatic pilot according to manufacturer's instructions. Use thermostat to control oven heat. Turn gently, do not force.

3. Valves. Keep all burner valves clean and well lubricated. Lubricate burner valves with valve lubricant furnished with oven at least once

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every three months. For additional supply of valve lubricant, requisition Grease, Plug, Valve, Silicon Base, NSN 6850-00-294-0860.

4. Load shelves evenly, making sure food does not contact oven linings.

5. Do not overload oven compartment.

6. Clean oven daily or if necessary more often. Clean oven exterior and interior with damp cloth or mild detergent and wipe dry. Scrape all carbonized matter from door, door frame, and outside lower lip of door using a scraper. Clean shelves daily or if necessary more often.

3.8 Repair and maintenance. Provisions shall be made to allow ready adjustment, service, and replacement of gas system components and operational electrical accessories and components, without disturbing or disassembly of any other unrelated major component or equipment.

3.9 Finish. Ovens shall be smooth on all surfaces and edges and shall be free from discoloration or stains. Oven compartments shall be finished in accordance with the manufacturer's standard practice for stainless steel, aluminized steel, or porcelain enamel on carbon steel, as applicable.

3.10 Workmanship. All components and assemblies of the oven shall be free from dirt and extraneous material, burrs, slivers, rough die, gouge, tool, trim and grind marks, dents and cracks. Castings, and molded parts if used, shall be free of sand, fins, pits, blowholes, and sprues. External surfaces for all components and assemblies shall be free from sharp edges and sharp corners.

3.10.1 Metal fabrication. Metal used in fabrication of the oven shall be free from kinks. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies or fixtures to insure uniformity of size and shape.

3.10.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be smooth and free from cracks, burn holes, undercuts, or incompleted fusion. All scale and flux deposits (when flux is used) shall be removed from finish welds, and rough edges and projections shall be dressed smooth.

3.10.3 Fastening devices. Threaded fasteners and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked, or stripped and shall be securely

force.

3. Valves. Keep all burner valves clean and well lubricated. Lubricate burner valves with valve lubricant furnished with oven at least once

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 In-process examination. Examination shall be made of the following processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the affected items and process:

a. Outer shell stiffened and reinforced, when applicable (see 3.5.1).

b. Oven compartment lining(s) reinforced and without open joints (see 3.5.2).

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4.2.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of ovens of one type and style offered for acceptance at one time. The sample unit shall be one oven. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified.		X
	Enameled surfaces blistered, peeled, chipped, or area of no film.		X
	Evidence of discoloration or stains.	X	
Construction and workmanship	Component missing.	X	
	Any burr, sliver, sharp edge, or rough die, tool, gouge, trim or grind marks.	X	
	Any component fractured, buckled, dented, punctured, or malformed.	X	
Body	Not constructed as specified.	X	
	Not fabricated of material specified. (Where visual determination can be made.)	X	
Oven top	Not as specified.	X	
	Not provided with stainless steel top.	X	
Oven compartment	Evidence of open joints in linings.	X	
Doors	Not full width of the oven interior.	X	
	Door windows (when required) do not permit viewing of the center three shelves.	X	
	Door handle missing.	X	
	Doors when fully open obstruct removal of shelf or shelf supports.	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Shelf and shelf support (as applicable)	Not removable.	X	
	Not furnished with a minimum of five shelves.	X	
	Shelf jams or buckles when inserted in or withdrawn from shelf support.		X
	Shelf support not easily removable.		X
Vent control or air baffle	Vent control or air baffle not as specified (when provided).	X	
Gas burners	Not removable from front of oven.	X	
	Not protected or supported.	X	
	Oven blower not controlled by door switch (i.e.) does not shut off when door is open or turn on when door is closed.	X	
	Not located to prevent heat loss.	X	
	Not accessible for cleaning or servicing.	X	
	Gas burner valve orifice not fixed type.	X	
	Orifice not color coded.	X	
Lights	Not accessible for replacing.	X	
	Exterior light control missing.	X	
Interior lighting	Not recessed or protected in compartment.	X	
	Not located to provide maximum illumination.	X	
Indicator light	Does not indicate when component is energized.	X	
Thermostat	Improper range.	X	
	Not protected from direct heat.	X	
	Not protected from grease and moisture.	X	
Timer	Not provided with an alarm signal.	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Open or enclosed stand or legs (as applicable)	Open or enclosed stand extends beyond oven sides and front.		X
	Open or enclosed stand not as specified.	X	
	Open or enclosed stand not reinforced.	X	
	Legs not reinforced (when secured directly to oven bottom).	X	
	Leveling devices missing.	X	
Flexible gas line	Not type specified.	X	
Connector	Not provided.	X	
Welding (when applicable)	Not free of oxide, scale, paint, and grease.	X	
	Not continuous, sound, smooth, or free from porosity, cracks, incomplete fusion, and deformation of material.	X	
	Flux (when used) not removed.		X
Threaded fasteners and rivets	Missing, broken, stripped, or fractured.	X	
	Loose.		X
	Rivets not fully peened or upset.		X
	Rivets head(s) not in full contact with riveted members.		X
Data-name plate	Omitted or not as specified.		X
	Information incomplete or not legible.		X
	Not located so as to be readily visible to operator.		X
User's instruction plate	Omitted or not as specified.	X	
	Information incomplete or not legible	X	
	Not permanently fastened to outside front of unit.	X	

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4.2.4 End item dimensional examination. The end item shall be examined for compliance with the dimensions specified in table I. Any dimension that is not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of ovens of one type and style offered for acceptance at one time. The sample unit shall be one oven. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.5 End item testing.

4.2.5.1 Representative item testing. One representative end item from the lot shall be tested as specified in 4.4.3, 4.4.4, and 4.4.5. Any test failure shall be cause for rejection of the lot.

4.2.5.2 Operational testing. Every end item shall be tested as specified in 4.4.2. Any end item exhibiting one or more test failures shall be rejected.

4.2.5.3 Shelf strength testing. The end items shall be tested for shelf strength as specified in 4.4.1. The lot size shall be expressed in units of ovens. The sample unit shall be one oven. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for test failures.

4.2.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-B-2316.

4.3 Code and standard compliance. Certifications, certified test reports, or listing marks for standards submitted in accordance with 3.2 shall be examined and validated as proof of compliance. Prior to Government approval of first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the convection ovens he proposes to furnish under this document, meet the applicable requirements of NSF Standard No. 4 and ANSI Standards Z 21.41, Z 21.45, and Z 83.12 as follows:

4.3.1 NSF. Acceptable evidence of meeting the requirements of NSF Standard No. 4 shall be one of the following.

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished convection oven, or

b. A certification for the convection ovens issued by NSF under their special one-time contract evaluation/certification service, or

c. A certified test report, acceptable to the contracting officer with the advice of the Army Surgeon General, from an independent testing laboratory indicting that the convection ovens have been tested and conform to the specified NSF Standard.

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4.3.2 ANSI certification. Acceptable evidence of meeting the applicable requirements of Z 21.41, Z 21.45, and Z 83.12 shall be a reproduced copy of the American Gas Association (A.G.A.) Appliance Certificate, a listing in the latest A.G.A. "Directory of Certified Appliances and Accessories", or a certificate test report from a recognized independent testing laboratory, acceptable to the Government, indicating compliance with ANSI Standards Z 21.41, Z 21,45, and Z 83.12.

4.4 Methods of inspection.

4.4.1 Shelf structural strength test. The shelf shall be held in its normal position in the oven and subjected to a uniformly distributed load of 20 pounds per square foot. The shelf shall be examined for deflection and dislodgement. After 2 minutes, the load shall be removed and the shelf examined for permanent deflection. Inability of the shelf to meet the load requirements specified in 3.5.4 shall constitute failure of this test.

4.4.2 Operational test. Each oven compartment shall be connected to the specified gas supply and all components and connections tested with liquid soap solution to determine compliance with the gas leakage requirements specified in 3.5.7.4. Burners shall be tested by ignition to determine that they function properly and controls and alarm signal shall be tested by turning off and on to determine compliance with the operational requirements of 3.5.7.4. Any leakage or any nonconformance with the operational requirements specified shall constitute failure of this test.

4.4.3 Oven thermostat accuracy test. Each oven compartment shall be turned on, each thermostat shall be successively set at 250°F, 300°F, and 400°F, and the compartment temperature shall be measured to determine compliance with temperature requirements specified in 3.5.8. Inability to meet the thermostat accuracy requirements specified shall constitute failure of this test.

4.4.4 Preheat time with blower energized test. Prior to testing the oven, the ambient temperature shall be 70°F plus or minus 5°F with the door closed. The thermostat for one compartment shall be set to at least 400°F and the convection oven blower energized to the manufacturer, recommended setting. The compartment temperature shall be read at 1 minute intervals. The time required to reach a temperature of 400°F shall be recorded. When applicable, the test shall be repeated for the second compartment of type II ovens to determine compliance. Inability to meet the oven temperature requirements specified in 3.5.8.1 shall constitute failure of this test.

4.4.5 Circulation and heat distribution test. This test shall be performed in conjunction with the preheat time test specified in 4.4.4. Thermocouples shall be located on the top, middle, and bottom shelves in the following manner: Lines shall be drawn on each of the specified shelves, 2 inches in and parallel to the front, back, and each side; mid-point to mid-point of each side; and mid-point of the back to mid-point of front. A thermocouple shall be located at each line intersection for a total of nine per shelf. All

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temperatures shall be read on a recording potentiometer. Use of deflection type instruments shall not be considered satisfactory. At the thermostat setting used in 4.4.4, all temperatures on each shelf in the oven compartment shall be recorded. The mean temperature on each shelf shall then be calculated. Any nonconformance with the circulation and heat distribution requirements specified in 3.5.7.1 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Level A, B, and C. Preservation, packaging, packing and marking for each level of protection shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use. The baking and roasting gas convection ovens are for use in kitchens and bakeries where combined use and time savings are required.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and style of oven required (see 1.2).
c. When a vent control or air baffle is required (see 3.5.2.1).
d. When aluminized-coated steel or porcelain enamel on carbon steel for the door interior is required (see 3.5.3).
e. Type of gas required (see 3.5.6).
f. Type of liquified petroleum gas (LPG) required (see 3.5.6.2).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.4 Supersession information. Since this document now covers only stationary ovens with removable supports and removable shelves, the designations "Style 1 - Stationary" and "Model B - With removable shelf supports and removable shelves" have been deleted.

6.5 Changes from previous issue. Asterisks are not used in this revision to identity changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 7310-0696

Review activities:

Army - MD, TS
Navy - MC
Air Force - 84
DLA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-43679D	OVENS, BAKING AND ROASTING, FORCED CONVECTION, GAS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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